Worked Problems with Equations Pasted as Pictures

Fractions Inside Sentences

Fractions that sit inside a sentence were pasted from an equation editor as small pictures.

The mean of 18 equal parts of a total T is $\frac{T}{18}$.

Dividing both sides by 7 gives $x = \frac{9}{7}$.

The common denominator of $\frac{1}{5}$ and $\frac{1}{2}$ is their product when the numbers share no factor.

Multiplying $\frac{x}{7}$ by 7 removes the denominator and leaves the single letter x.

To compare $\frac{11}{14}$ with $\frac{11}{2}$, cross-multiply and compare the two products.

The average rate of change over the interval is $\frac{f(5)-f(11)}{5-11}$.

Two pipes together fill $\frac{1}{5}+\frac{1}{3}$ of the pool in one hour.

Determinants and Matrices

Arrays of numbers were pasted the same way, one picture for each display.

The determinant of a two by two matrix is the product of the diagonal entries minus the product of the others:

$$\det A = \begin{vmatrix} -1 & -5 \\ 7 & 7 \end{vmatrix} = 28 \tag{2.1}$$

Expanding along the first row gives the determinant of a three by three matrix:

$$\det B = \begin{vmatrix} 3 & 9 & 15 \\ 5 & 9 & 9 \\ 2 & -5 & 9 \end{vmatrix} = -510$$

Scaling entries scales the terms of the determinant:

$$\begin{vmatrix} 3a & b \\ 7c & d \end{vmatrix} = 3ad - 7bc$$

The characteristic condition below is a quadratic equation in x:

$$\begin{vmatrix} x - 13 & 11 \\ 1 & x + 13 \end{vmatrix} = 0 \tag{2.2}$$

Cramer's rule writes each unknown as a ratio of determinants:

$$x = \frac{\begin{vmatrix} 5 & 11 \\ 3 & -5 \end{vmatrix}}{\begin{vmatrix} -7 & 11 \\ 9 & -5 \end{vmatrix}}, \quad y = \frac{\begin{vmatrix} -7 & 5 \\ 9 & 3 \end{vmatrix}}{\begin{vmatrix} -7 & 11 \\ 9 & -5 \end{vmatrix}}$$

Vertical bars around an array also denote the determinant:

$$\begin{vmatrix} 9 & 15 \\ 0 & 5 \end{vmatrix} = 45$$

Transposing a matrix swaps its rows and columns:

$$A = \begin{bmatrix} -9 & 7 & -4 \\ 7 & 9 & 7 \end{bmatrix}, \qquad A^T = \begin{bmatrix} -9 & 7 \\ 7 & 9 \\ -4 & 7 \end{bmatrix}$$

A matrix times a column vector gives another column vector:

$$\begin{pmatrix} -3 & -1 \\ 9 & -9 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} -13 \\ -9 \end{pmatrix} \tag{2.3}$$

The inverse of a two by two matrix uses the determinant and swaps entries:

$$A^{-1} = \frac{1}{29} \begin{bmatrix} -7 & 11 \\ -2 & -1 \end{bmatrix}$$

The augmented matrix of a system keeps the right-hand side beside the coefficients:

$$\left[\begin{array}{cc|c} -13 & 3 & 11 \\ 15 & 2 & 12 \end{array}\right]$$

A three by three system gives an augmented matrix with one extra column:

$$\left[\begin{array}{ccc|c} -1 & 1 & -15 & 15 \\ 3 & -5 & 15 & 15 \\ 7 & 1 & -7 & 19 \end{array}\right] \tag{2.4}$$

Row operations turn the left block into the identity and read off the solution:

$$\left[\begin{array}{cc|c} 3 & 7 & -6 \\ -3 & -2 & 6 \end{array}\right] \longrightarrow \left[\begin{array}{cc|c} 1 & 0 & -2 \\ 0 & 1 & 0 \end{array}\right]$$

Systems and Cases

Braces and cases are pictures too, so the text layer holds none of their symbols.

Solve the system of equations by elimination:

$$\begin{cases} -9x + 2y = -32 \\ 7x + 11y = 50 \end{cases}$$

Find every pair of numbers that satisfies the system:

$$\begin{cases} -5x + 5y = 0 \\ 3x + 5y = 16 \end{cases} \tag{3.1}$$

Use substitution to solve the following system:

$$\begin{cases} 3x + 5y = 14 \\ 5x - 9y = 6 \end{cases}$$

Determine the values of the unknowns in the system:

$$\begin{cases} -3x + 11y + 3z = -11 \\ -5x + 7y + 9z = -11 \\ -5x - 3y + 3z = 5 \end{cases}$$

A function is defined in pieces:

$$h(x) = \begin{cases} x + 7 & \text{when } x \geq 0 \\ 7 - 7x & \text{when } x < 0 \end{cases}$$

A function is given by cases:

$$h(x) = \begin{cases} x + 9 & \text{when } x \geq 0 \\ 9 - 3x & \text{when } x < 0 \end{cases}$$

A function is defined in pieces:

$$g(t) = \begin{cases} 9t & \text{for } 0 \leq t \leq 9 \\ 81 & \text{for } 9 < t \leq 18 \\ 81 - 7(t - 18) & \text{for } t > 18 \end{cases} \tag{3.2}$$

A function is given by cases:

$$g(t) = \begin{cases} 7t & \text{for } 0 \leq t \leq 5 \\ 35 & \text{for } 5 < t \leq 16 \\ 35 - 2(t - 16) & \text{for } t > 16 \end{cases}$$

Derivations and Other Displays

The last group holds longer displays, some of them numbered at the right margin.

Isolating the unknown takes three short steps:

$$\begin{aligned} 2x + 5 &= 23 \\ 2x &= 18 \\ x &= 9 \end{aligned} \tag{4.1}$$

Completing the square solves the quadratic in four steps:

$$\begin{aligned} x^2 + 10x + 13 &= 0 \\ (x+5)^2 &= 12 \\ x + 5 &= \pm\sqrt{12} \\ x &= -5 \pm \sqrt{12} \end{aligned}$$

The power rule is applied term by term:

$$\begin{aligned} \frac{d}{dx}\left(5x^3 + 15x\right) &= 5 \cdot 3x^2 + 15 \\ &= 15x^2 + 15 \end{aligned}$$

Matching bases turns an exponential equation into a linear one:

$$\begin{aligned} 2^{x+2} &= 32 \\ 2^{x+2} &= 2^5 \\ x + 2 &= 5 \\ x &= 3 \end{aligned}$$

A further result is displayed below:

$$\int_{0}^{2} (9x + 1)\,dx$$

A further result is displayed below:

$$\sum_{k=1}^{10} (5k + 1)$$

A further result is displayed below:

$$\frac{d}{dx}\left(2x^3 - 9x^2\right)$$

A further result is displayed below:

$$\sin\theta = \frac{11}{13}$$

Summary of Results

Quantity	Value	Unit
Distance	12	m
Acceleration	1.5	m/s^2
Density	0.072	g/cm^3
Area	44	cm^2

Handout Details

Return the sheet. Hand in the answers on Friday at the start of class.

Questions. Ask during office hours or write them on the board before the lecture.